

AUGIE RODRIGUES

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EDUCATION

Rutgers University, New Brunswick

Sept 2022 - Dec 2024

BS in Computer Science, Cum Laude (GPA: 3.51/4.00)

- **Coursework:** Computer Architecture, Software Methodology, Design & Analysis of Computer Algorithms, Data Structures, Operating Systems Programming, Linear Algebra, Principles of Programming Languages, Discrete Structures I & II

SKILLS

Programming: Java, C, C++, Python, C#, TypeScript, SQL, Matlab, Rust, Go, Bash scripting, x86 Assembly

Frameworks & Tools: Git, GitHub, Linux, GNU Make, GDB, Docker, Node.js, React, Spring Boot, .NET, SQLite, REST APIs

EXPERIENCE

Online Orders Specialist, Stop & Shop

May 2023 - present

- Restored online order access by resolving software outages, preventing cancellations, and recovering over \$10,000 in sales
- Verified product availability and updated inventory records at store opening and closing to maintain accurate stock levels

Substitute Teacher, Caldwell-West Caldwell Public Schools

Dec 2025 - present

- Demonstrated punctuality and flexibility in accepting short-notice assignments via phone and digital communication channels
- Provided academic support to struggling students to improve subject comprehension, problem-solving skills, and assignment completion

Server's Assistant, Mountain Ridge Country Club

May 2022 - Aug 2022

- Prepared dining areas for service by setting up indoor and outdoor seating and arranging table settings
- Supported club events, maintaining hospitality standards while handling last-minute table adjustments and assisting guests

Freelance Web Developer, Self-Employed

Feb 2021 - present

- Implemented e-commerce functionality, Stripe & PayPal payment processing, and responsive HTML/CSS/JS interfaces
- Managed deployment and maintenance through Git-based version control and domain/DNS configuration

PROJECTS

CHIP-8 Emulator

Nov 2025 - Mar 2026

Cycle-Accurate CHIP-8 Interpreter with SDL2 Rendering and Full Opcode Implementation

- Designed a cycle-accurate CHIP-8 virtual machine in C with full CPU state management and 4KB memory model
- Implemented complete instruction set (35 opcodes), including arithmetic, bitwise logic, branching, timers, and input handling
- Built 64×32 XOR-based sprite renderer with bitwise pixel compositing, VF collision flag detection, and framebuffer updates
- Integrated SDL2 for real-time display scaling, keyboard input polling, and sound-timer-driven audio via host event loop
- Enforced deterministic fetch-decode-execute timing with configurable cycle rate and accurate delay/sound timer semantics

Custom UNIX-Like Shell

Feb 2024 - May 2024

POSIX-Style Shell with Fork/Exec, Pipelining, and File Descriptor Management

- Developed a POSIX-style shell in C using GCC and GNU Make with modular process and I/O abstractions
- Enabled advanced shell features such as piping and command chaining, replicating commands like **grep**, **ls**, and **cat** pipelines
- Incorporated multithreading using **fork** to handle concurrent processes, allowing the shell to execute commands in parallel
- Implemented terminal and file I/O handling via **fopen**, **read**, and **write**, including support for input/output redirection
- Managed file descriptor duplication using **dup2** to enable input/output redirection and pipe orchestration

BookLogger

July 2025 - Oct 2025

Full-Stack .NET Desktop Application with Entity Framework and Secure Identity Management

- Created a WPF application in C# on .NET with XAML-based GUIs, employing the MVVM pattern and data binding
- Employed Entity Framework Core with SQLite backend for caching, using LINQ queries to display user and book data
- Implemented authentication and user management with secure password hashing and validation via ASP.NET Core Identity
- Retrieved book metadata from an external API and implemented caching to efficiently populate search results

Restaurant Queue & Order Management System

Oct 2024 - Dec 2024

Event-Driven Order Management Application with Custom Collections and MVC Design

- Built a restaurant management app with menu and order handling, applying Java, MVC architecture, and OOP principles
- Implemented core classes using enums, abstract classes, and interfaces to model restaurant operations and user orders
- Managed orders and menu items using custom Lists and Heaps to efficiently process and access order data
- Designed GUIs with FXML and CSS, linking Android components to backend logic via Adapters and view bindings